

Tutorial 8 - User-defined Types

1. The table below shows the product numbers and their respective unit prices.

Product Number	Unit Price
7	345.00
8	853.00
9	471.00
10	933.00

Use *typedef* to rename the *int* type to **ProductNo**. Then declare a variable named **pNo** and initialize it with value 7. Write a *switch* statement to assign the unit price to a variable named **uPrice** according to the value of **pNo**.

2. You are building a weather monitoring system where temperatures are always stored as double values. To make the program more readable and easier to maintain, you decide to give this data type a meaningful alias.

Use *typedef* to define a new type named **Temperature** for double. Then, write a C++ program that declares and initializes two variables of type **Temperature** with morning and afternoon readings (in celsius), calculates their average, and displays it.

3. Write a C++ program that prompts the user to enter their **first name**, **last name**, and full **address** details (street, city, state, postal code). Use *typedef* to define all name and address components as custom string types (e.g., **FirstName**, **Street**, etc.). After collecting the inputs from the user, display the user's name in the format "**Last, First**" followed by the full, properly formatted address.

4. You are creating a hospital management system. Each patient's record should include: ID, name, age, blood group, and bill amount. Since this structure will be used often, create a **patientRecord** using *typedef* for easier reference.

- a) Write a C++ main function that declares a variable **patient** of **patientRecord**, initializes it with some realistic values, and displays the patient details except the bill amount. Then it will perform the function in part (b) below.
- b) Write a C++ function **applyDiscount** that takes the information of **patientRecord** type, it will check and give a 20% discount on the bill amount if the patient is 60 years or older. Finally, it prints the final bill after the discount or no discount is given.

5. What is wrong with the following pair of enumeration type declarations

```
enum Colors {RED, ORANGE, YELLOW, GREEN, BLUE, INDIGO, VIOLET};  
enum Flowers {ROSE, DAFFODIL, LILY, VIOLET, COSMOS, ORCHID};
```

6. Use an enumerated type to represent the planet listed below.
Mercury, Venus, Earth, Moon, Mars, Jupiter, Saturn, Uranus, Neptune, Pluto

7. A coffee shop sells three types of coffee: **Espresso, Latte, and Cappuccino**. Each has a fixed price:

- Espresso = RM 8
- Latte = RM 10
- Cappuccino = RM 12

Write a C++ program using an *enum* to represent the coffee types. Then it should perform the following tasks:

- a) Display a Coffee Menu for these 3 types of coffee
- b) Prompt user to select one type of coffee from the menu
- c) Prompt user for the quantity required
- d) Calculate and display the total price payable

Below is the sample output screen:

```
=== Coffee Shop Menu ===
1. Espresso (RM 8)
2. Latte (RM 10)
3. Cappuccino (RM 12)
Choose your coffee (1-3): 2
Enter quantity: 2

====> Total price = RM 20
```

8. Simulate the traffic light system using *enum* where the states of traffic light and their actions are:-

RED->STOP, GREEN->GO, YELLOW->SLOW DOWN

Write a C++ program to prompt the user to select a traffic light (1=RED, 2=YELLOW, 3=GREEN), then display the action of the traffic light selected. Note you are required to use *typedef* to represent the integer value as light to be selected by the user.

9. Cinema tickets have different prices depending on **age groups** as below:

- CHILD (below 12) = RM 8
- TEEN (12–17) = RM 12
- ADULT (18–59) = RM 20
- SENIOR (60+) = RM 10

Use an *enum* called **AgeGroup** to represent the age groups and write a C++ function named **classifyGroup** which will return the corresponding age group based on the different range of age given above.

Develop another C++ function **getPrice** to receive the age group and return the correct price accordingly.

In the **main** function, users will be prompted to enter an age, then determine the age group and display the corresponding price for the cinema ticket.

10. A car has different gear modes that control how it moves. The main gears are:

Mode	Gear	Action
1	PARK	Car stopped
2	REVERSE	Car moving backward
3	NEUTRAL	Car idle
4	DRIVE	Car moving forward

Write a C++ function **findGearAction** that receives the **gear** of an enumerated type **Gear** (represent these gear modes). It will return the description of the car's action as per the above table.

Create a **main** function to allow the user to select a gear by entering a mode number as in the table above, then it should print the correct action description of what the car will do in that gear.